# The title

Karolina Muszyńska[1] & To fill in later on[2]

[1] University of Warsaw, Faculty of Psychology

[2] To fill in later on

## Author Note

The authors made the following contributions. Karolina Muszyńska: Conceptualization, Writing - Original Draft Preparation, Writing - Review & Editing; To fill in later on: Writing - Review & Editing, Supervision.

Correspondence concerning this article should be addressed to Karolina Muszyńska, University of Warsaw, Faculty of Psychology, ul. Stefana Banacha 2D, 02-097 Warsaw, Poland. E-mail: karolina.muszynska@psych.uw.edu.pl

## Abstract

Abstract here


*Keywords:* keywords

Word count: X

# The title

## Methods

### Participants

In the American English validation study, the participants were 204 parents of children aged 15 to 36 months ($M = 26.21$, $Mdn = 26$, SD = 6.37). There were 103 girls in the sample (50.49%). The sample was ethnically diverse, including White, Black, Latino/Hispanic, Asian, Native American, and other races/ethnicities. None of the children were reported to hear other languages apart from English. The average number of years of education attained by the primary caregiver was 15.82 (SD = 2.27).

In the Polish validation study, the participants were 113 parents of children aged 18 to 34 months ($M = 24.73$, $Mdn = 24$, SD = 4.47). There were 49 girls in the sample (43.36%). The sample was White (reflecting the limited ethnic diversity within the Polish population). However, there were 28 children multilingual with Polish (as their home language) and some other (societal) language, living outside of Poland. Parents were asked about their education level and 65 reported having higher education or PhD, 1 reported having unfinished higher education, 11 reported having secondary education and 36 did not report their education level.

Notably, the two validation samples differed slightly (but significantly) in the mean age of children, $\Delta M = -1.49$, 95% CI $[-2.69, -0.28]$, $t(297.82) = -2.42$, $p = .016$, which may affect the outcomes of direct comparisons between the samples, where age could be a contributing factor. We present one such comparison–standard error of CAT ability–where this difference is explicitly discussed. *NOTE*: You can also view/comment on this on GitHub Issue #7 linked here.

## Materials

The data was gathered with CAT-CDIs in American English and Polish. The American English CAT-CDI was created from the items that were common to three CDIs: Words and Gestures, Words and Sentences and CDI-III. With these items, two CAT-CDIs were created, one for word production and another for word comprehension (see Kachergis et al. 2022 for more details). The Polish CAT-CDIs were created following a largely similar procedure as the ones in American English, but there was a separate CDI-CAT developed for each CDI version (CDI: Words and Gestures and CDI: Words and Sentences) and for each CDI:WG subscale (word production, word comprehension, gesture use) (Krajewski et al. in preparation). The American English CAT-CDI word production included 679 items and the Polish CAT-CDI version of WS: Words and Sentences included 666 items. There are 395 words common to both CDI-CAT language versions.

## Procedure

Though in both American and Polish validation studies the parents were asked to fill in both the full CDI and the CAT-CDI, the procedure differed between the two studies. In the validation of American English CDI-CAT, parents were given both CDIs in one sitting, but the order of the CDI versions was counterbalanced. In the Polish validation study, the order of the CDIs was fixed, with the full CDI always first, and then, after 1–30 days ($M$ = 8.31, $Mdn$ = 8, SD = 4.90) parents were asked to fill in the CAT-CDI. Both studies were done fully online and obtained the approval of adequate ethics committees.

# Results

## Psychometric properties of the two CAT-CDIs

Our first aim was to examine whether CAT-CDIs in American English and Polish demonstrate comparable psychometric properties. To that end, we revisit the psychometric properties reported for the American English CAT-CDI (word production) in Kachergis et al. (2022) and compare those to the data from Polish CAT-CDI (Words and Sentences).

We found similarly strong correlations in the two languages between the abilities estimated from CDI-CAT and full CDI scores (English and Polish: $r$ = .86). The correlations were also strong within individual age groups (see Tables 1 and 2). We also obtained ability estimates from the full CDIs, which is a robust measure of ability because it is based on all available item responses, maximizing precision. Such ability from the full test version is often considered a baseline for comparisons with ability estimated from a short (adaptive) version. We correlated the abilities estimated from the CDI-CAT and abilities estimated from full CDI and again found strong correlations in both languages, English and Polish: $r$ = .92. Finally, we also correlated the abilities estimated from the full CDI and the full CDI scores, i.e. two measures obtained from the same data. This way we examine consistency between model-based abilities and traditional scores. Again, we found strong correlations in both languages (English: $r$ = .95, Polish: $r$ = 0.94), confirming that both measures reflect the same underlying ability.

Table 1

*English: Correlations between ability estimated by CAT-CDI and ability estimated from full CDI by children's age.*

| | [15,18) | [18,21) | [21,24) | [24,27) | [27,30) | [30,33) | [33,36] |
|---|---|---|---|---|---|---|---|
| r ability CAT vs full CDI | 0.95 | 0.85 | 0.82 | 0.83 | 0.59 | 0.84 | 0.86 |
| N | 26 | 22 | 26 | 30 | 28 | 24 | 48 |

Table 2

*Polish: Correlations between ability estimated by CAT-CDI and ability estimated from full CDI by children's age.*

| | [15,18) | [18,21) | [21,24) | [24,27) | [27,30) | [30,33) | [33,36] |
|---|---|---|---|---|---|---|---|
| r ability CAT vs full CDI | NA | 0.8 | 0.94 | 0.91 | 0.89 | 0.95 | NA |
| N | NA | 29 | 22 | 16 | 23 | 22 | 1 |

We also looked at the mean squared error (MSE) which quantifies the differences between the abilities as estimated by CAT-CDI and from the full CDI (the error is squared to eliminate negative values). The mean squared error in English was 0.55 ($Mdn$ = 0.17, $SD$ = 1), and in Polish it was 0.19 ($Mdn$ = 0.08, $SD$ = 0.45). The mean squared error was relatively high in English (compared to Polish), suggesting a relatively larger discrepancy between CAT-CDI ability estimates and full CDI ability estimates in English. This occurred despite strong correlations between the two measures. This difference may be explained by at least two reasons. First, a number of participants with extreme discrepancies between the CAT-CDI and full CDI estimates may be inflating the MSE.

Second, there might be a systematic bias, with one estimate consistently higher than the other.

To explore the first of these possibilities, we identified children for whom the estimates from the CAT-CDI and full CDI diverged extremely, i.e. their errors was removed by 1.5 SD from the mean. There were 15 such cases (7.35%) in the English dataset and 4 cases (1.96%) in the Polish dataset. We then re-calculated the mean squared error without the cases of extreme discrepancy, which yielded a MSE of 0.44 ($Mdn = 0.29$, $SD = 0.44$) in English and MSE of 0.12 ($Mdn = 0.07$, $SD = 0.14$) in Polish. The slightly reduced MSE values indicate that extreme discrepancies had only a modest impact on the overall error and do not fully explain the MSE values and the differences observed between ability estimates from full CDI and CAT-CDI.

Next, we examined a possible bias in our two ability estimates. We found that the mean CAT ability estimates were consistently higher than the full CDI ability estimates. That was true both in English, $M_D = 0.55$, 95% CI $[0.48, 0.62]$, $t(203) = 15.11$, $p < .001$ (large effect size: Cohen's $d = 1.06$, 95% CI $[0.89, 1.23]$), and in Polish, $M_D = 0.26$, 95% CI $[0.19, 0.32]$, $t(112) = 7.71$, $p < .001$ (medium effect size: Cohen's $d = 0.73$, 95% CI $[0.52, 0.93]$).

The question that remains is whether this consistent difference between the two estimates of abilities comes from CAT over-estimating child's lexical ability or from the full CDI underestimating child's ability. For one, it may be that in CAT-CDI parents answer "yes, my child produces this word" to more items that would be expected, and as a result, CAT-CDI may overestimate the child's true ability. On the other hand, it could be that as the full CDI is very long, parents may omit marking some items that their child produces (e.g. Łuniewska et al., 2025), which can result in the full CDI underestimating the child's true ability. This question is difficult to answer without direct assessment of child's lexical ability. However, if parents are prone to omitting items in the full CDI, this would be most visible in the case of children with large vocabularies (the more words the child knows, the more CDI items can be omitted). Indeed, we see that as a child's ability grows (as estimated from the full CDI), the size of the discrepancy between the ability estimate from full CDI and the CAT CDI grows as well, English: $r = .72$, 95% CI $[.65, .78]$, $t(202) = 14.74$, $p < .001$, Polish: $r = .28$, 95% CI $[.10, .44]$, $t(111) = 3.07$, $p = .003$.

[Here not finished, thinking of ways of testing out that CAT overestimates the ability] However, if CAT-CDI invites parents to respond "yes" more often than expected, we should be seeing… more "yes" responses the more items are being shown? or maybe simply many participants with more than half of their responses being yeses? or maybe we can get some THIRD value (not from the validation study) that we could compare these two validation ability estimates to?

## Item properties in the two CAT-CDIs

Our second aim was to analyze similarities and differences in IRT item properties and item selection in CAT in English and Polish.

There are 679 items in the English CAT-CDI and 666 items in the Polish CAT-CDI. In both languages, the items' difficulty and discrimination parameters were calculated using IRT 2 parameter model (these included separate samples, see Kachergis et al. 2022 and Krajewski et al. (in preparation)). An item's difficulty indicates the ability level at which there is a 50% probability that a participant will respond correctly. *NOTE*: See GitHub Issue 1. It is on the same scale as ability with negative values indicating difficult items, values around 0 indicating medium difficulty, and positive values indicating easy items. An item's discrimination indicates how well it distinguishes between individuals with slightly different ability levels–especially those near that difficulty point. Of these two parameters, item difficulty is of greater interest to the present paper as it is directly linked to ability and as discrimination power is more about how good the item is at measuring, rather than what it is measuring.

English items are more difficult than the Polish items, $\Delta M = -1.87$, 95% CI $[-2.06, -1.69]$, $t(1227.61) = -20.09$, $p < .001$ (English: min = -7.16, max = 4.45, $M$ = -2.19, $Mdn$ = -2.21, $SD$ = 1.98; Polish: min = -4.34, max = 4.41, $M$ = -0.32, $Mdn$ = -0.43, $SD$ = 1.41). This was true also for a subset of 390 items common to both languages - English items still proved to be more difficult than the Polish ones: $M_D = -1.68$, 95% CI $[-1.82, -1.54]$, $t(389) = -23.11$, $p < .001$. This is a result of the characteristics of the samples used to estimate the item parameters. In English, item difficulty was calculated based on a broader sample of children aged 12–36 months (spanning the CDI:WG, CDI:WS, and CDI-III), whereas the Polish data came from a sample of narrower age range of 18–36 months, corresponding to the CDI:WS. As a result, item difficulty in English was estimated using a relatively younger sample, for whom certain items may have been more challenging—thus appearing more difficult—compared to the older Polish sample. Still the item difficulty for common items in the two languages was positively and moderately correlated: $r = .65$, 95% CI $[.58, .70]$, $t(388) = 16.68$, $p < .001$.

Table 3

| category_pl | rho | n |
|---|---|---|
| quantifiers | 0.11 | 10 |
| clothing | 0.22 | 19 |
| connecting_words | 0.23 | 8 |
| locations | 0.31 | 6 |
| pronouns_demonstrative | 0.32 | 4 |
| places | 0.32 | 7 |
| vehicles | 0.45 | 9 |
| furniture_rooms | 0.47 | 17 |
| action_words | 0.50 | 75 |
| body_parts | 0.52 | 17 |
| prepositions | 0.55 | 10 |
| games_routines | 0.57 | 12 |
| time_words | 0.57 | 8 |
| household | 0.57 | 30 |
| outside | 0.58 | 20 |
| sounds | 0.60 | 6 |
| food_drink | 0.61 | 39 |
| descriptive_words (adjectives) | 0.62 | 19 |
| people | 0.67 | 16 |
| animals | 0.68 | 29 |
| toys | 0.74 | 15 |
| descriptive_words (adverbs) | 0.80 | 4 |
| question_words | 0.91 | 10 |

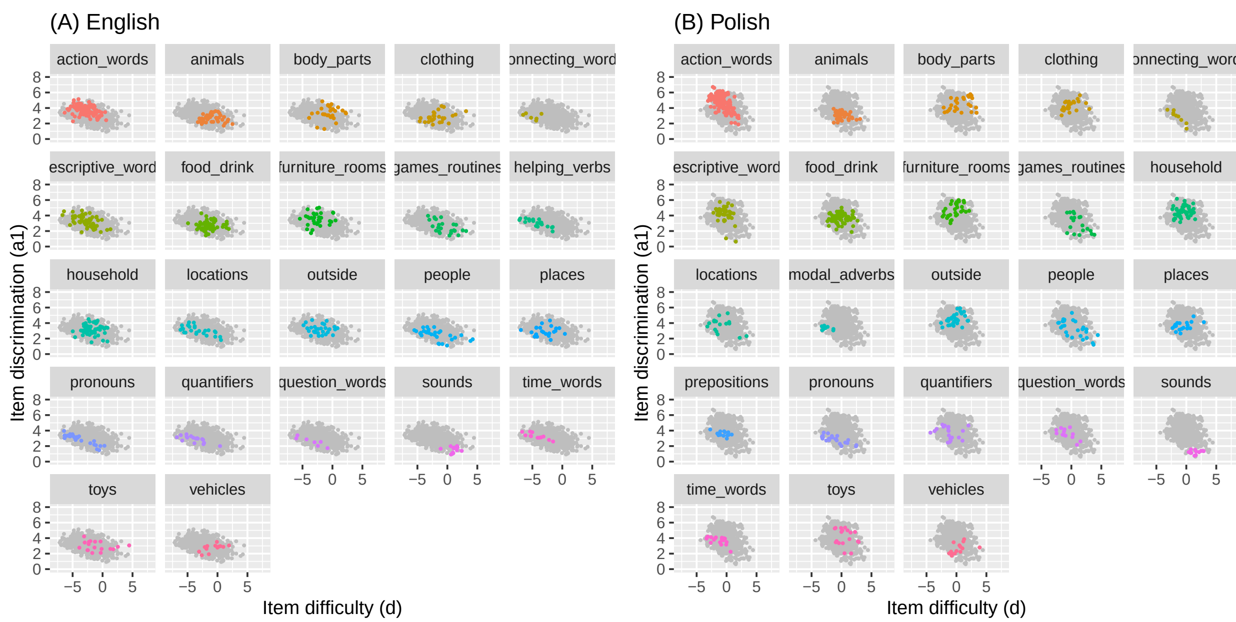


*Figure 1*. The relative positioning of the items by semantic category (colored) plotted in the context of the full item pool (grey): (A) English, (B) Polish.

We also wanted to do check whether items in particular CDI semantic categories show related difficulty values across the two languages. We performed a series of Spearman's rank correlations (on the difficulty of items common to CDI-CATs in Polish and English) for each CDI semantic category (see Table @ref(tab:d_by_cdi_category_tab)). The rank correlations coefficients vary by category, but for half of the categories the correlations are moderate to strong (0.52 to 0.91). Figure 1 shows the relative positioning of the items by semantic category (colored) plotted in the context of the full item pool (grey) in the two languages. It can be seen from the figure that many categories (e.g., sounds) show a similar distribution of items relative to the whole item pool across Polish and English.

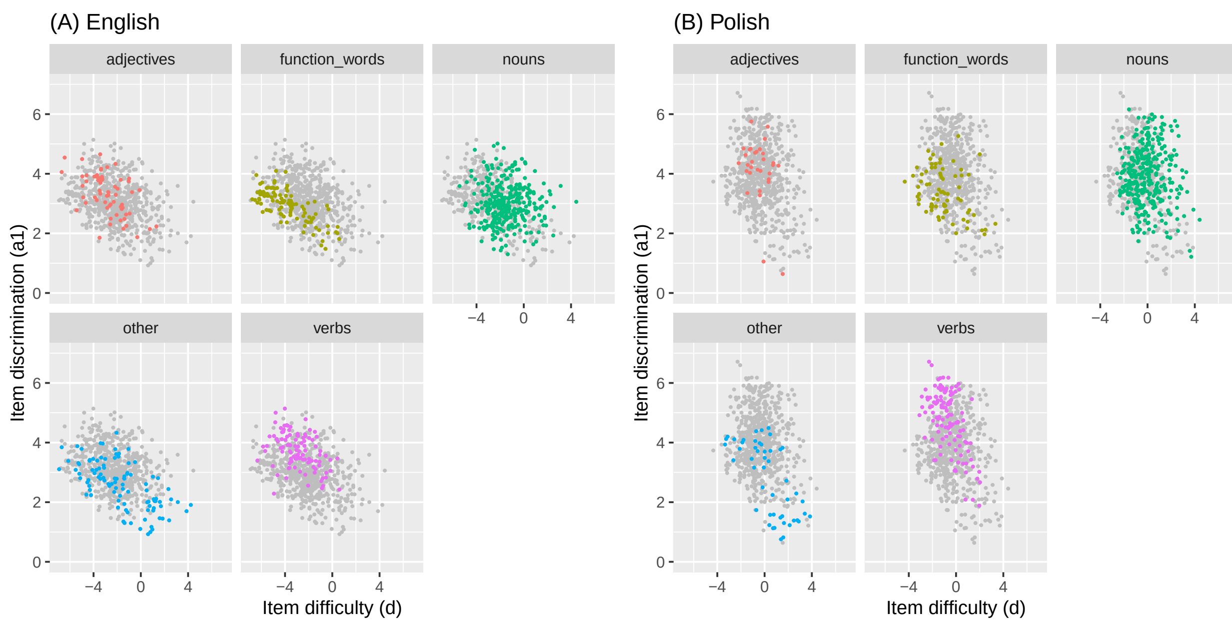


*Figure 2*. The relative positioning of the items by lexical class (colored) plotted in the context of the full item pool (grey): (A) English, (B) Polish.

Table 4

| lexical_class_en | rho | n |
|---|---|---|
| function_words | 0.40 | 50 |
| verbs | 0.49 | 71 |
| adjectives | 0.53 | 25 |
| nouns | 0.59 | 194 |
| other | 0.75 | 50 |

Last, we investigated whether particular lexical categories show related difficulty values across the two languages. Again, we performed a series of Spearman's rank correlations (on the difficulty of items common to CDI-CATs in Polish and English) for each lexical class (see Table @ref(tab:d_by_lexical_class_tab), Figure 2). The lowest correlation was for function words (0.40), then verbs (0.47), adjectives (0.53), nouns (0.6), other, i.e. mostly sounds and words relating to routines (e.g. “hello”, “yes”, “thank you”) and time (e.g. “tommorow”, “day”) (0.75).

## Item selection in the two CAT-CDIs

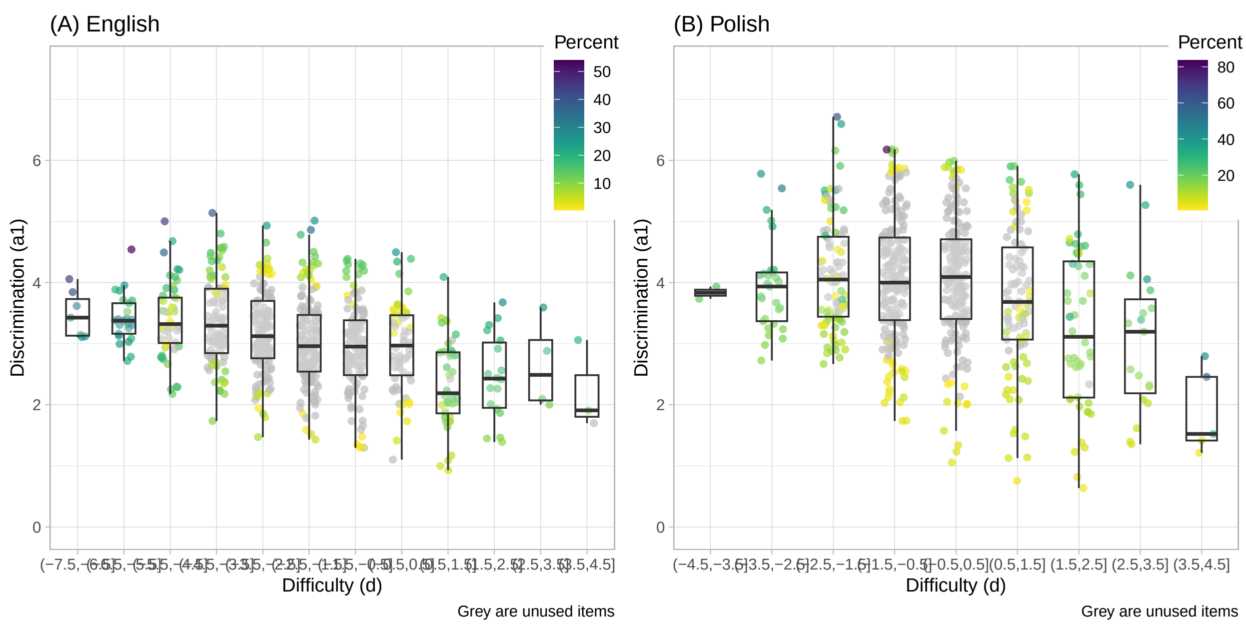


*Figure 3*. How often a given item was used in CAT-CDI administrations in the validation study in (A) English and (B) Polish. The items (points) are colored by the percentage of their appearance in the CAT-CDI administrations. Items colored in grey are items never used in any of the CAT-CDI administrations.

It is to be expected that a CDI-CAT will not need to administer all the items in the item bank. In fact, in the validation study the English CAT-CDI used 251 items (36.91%) and similarly, the Polish CAT-CDI used 258 items (38.74%). By design, a CDI-CAT selects items that are most informative for each participant. This means it draws from a subset of available items–typically those matched to the participant’s current ability estimate in terms of difficulty, and with high discrimination, meaning they effectively distinguish between individuals with abilities close to that estimate. Figure 3 plots the items by their difficulty and discrimination parameters and colors the items (points) by how often they were used in CAT-CDI administrations in English and Polish. A few findings are of note here. First, both CAT-CDIs used items of very low discrimination (i.e, items that do no discriminate well between ability levels). Often that was because for a given difficulty level there were not enough items and CAT-CDI had to use all the items available (this is particularly true of the items on the two ends of difficulty). However, more surprisingly, it is also true of items of medium difficulty, where items of higher discrimination were available (but were not chosen by the CAT algorithm). *NOTE*: See Issue 2 here.

Finally, some items keep being shown in many CAT administrations. In English, 3 items appear in more than half of administrations - *"long"* in 64% of administrations, *"make"* in 54% of administrations, and *"last"* in 53% of administrations. They are also high discrimination items in their difficulty bins. "Long" is additionally one of the starting items, i.e. a first item shown to the parent, chosen so that there is high probability the parent can respond positively. In Polish, 3 items appear in more than half of administrations - *"szukać"* (to look for) appearing in 83% of administrations, *"znaleźć"* (to find) appearing in 61% of administrations, and *"babcia"* (grandmother) appearing in 60% of administrations. All three items show very high discrimination. "Szukać" (to look for) and "znaleźć" (to find) are also in top 5 discrimination items in general. "Babcia" (grandmother) is one of the starting items.

To check whether the CDI items common in both languages may show similar frequencies in CAT-CDI administrations, we ran Spearman's rank correlation and found that the items frequencies in administrations were weakly correlated, $r_{\mathrm{s}} = .37$, $S = 6,420,909.29$, $p < .001$. However, the two CAT-CDIs differed in their potential length, as the English CAT-CDI was set to administer 25–50 items and the Polish CAT-CDI could administer up to 75 items (no minimum was set, but in practice the minimum items administered were 13). As an item is more likely to appear in longer tests (simply due to test length), then frequency of the Polish items could be inflated. *NOTE*:Could it be that, as an item is more likely to appear in longer tests (simply due to test length), frequency of the Polish items could be inflated? Could it influence the item's rank? I could probably recalculate item frequency not per total number of administrations, but per total number of items administered? SEE ISSUE #3

## CAT-CDIs' usefullness in research and practice

One of the key metrics for CAT-CDI's usefulness is whether it can shorten the CDI administration compared to the full CDI, but still obtain a reliable estimate of child's lexical ability. We have found that CAT-CDIs were significantly shorter to administer: the English CAT-CDI median time was 342s (~5.7 minutes) (median full CDI time was ) and the Polish CAT-CDI median time was 117s (~1.95 minutes) (median full CDI time was 1240s (~20.67 minutes)). Even though the CAT-CDI in Polish seemed shorter (compared to English CAT-CDI), the number of items administered in the two language versions of CAT-CDI was comparable: English $M$ = 31.42, $Mdn$ = 25 (25–50 items), Polish $M$ = 34.71, $Mdn$ = 23 (13–75 items).

The two CAT-CDIs differed in their settings as to when to finish the administration. Specifically, the Polish CAT-CDI was set so that the SEM as low as 0.1 be reached (with as few items as possible), and if the SEM could not be lowered to or below 0.1, the CAT-CDI stopped after administering a maximum of 75 items. The English CAT-CDI on the other hand was set to administer a maximum of 50 items, but the administration could be stopped earlier if the SEM of 0.15 or lower was reached. In other words, the Polish CAT-CDI prioritized as low SEM as possible, while the English CAT-CDI prioritized shorter test length. This difference resulted in slightly lower mean SEM in Polish CAT-CDI, compared to English CAT-CDI $\Delta M = 0.06$, 95% CI $[0.05, 0.07]$, $t(257.25) = 11.36$, $p < .001$ but comparable mean number of items administered, $\Delta M = -3.29$, 95% CI $[-7.74, 1.16]$, $t(137.00) = -1.46$, $p = .147$. However, it should be noted that the Polish and American validation samples differed slightly in age

$\Delta M = -1.49$, 95% CI $[-2.69, -0.28]$, $t(297.82) = -2.42$, $p = .016$, and as age is weakly correlated with SEM ($r = .18$, 95% CI $[.04, .31]$, $t(202) = 2.60$, $p = .010$), this slight difference in age could feed the SEM difference in the two validation samples. *NOTE*: We could consider matching kids from the two samples one-to-one on age, to even out this potential effect of age difference on SEM, see Issue #8.

In both CAT-CDIs, we have been able to reliably estimate lexical ability for majority of children. Taking as the cut-off Makransky's suggestion to consider a SEM below 0.2 as acceptable, so defined reliable CAT estimate was gathered for 96.5 of the American participants, and 92.6 of the Polish participants. The remaining participants, i.e. those with higher standard errors of measurement (in other words, lower reliability of ability estimates) were typically those at the extremes of the ability scale–children with either relatively low or relatively high lexical skills (see Figure 4).

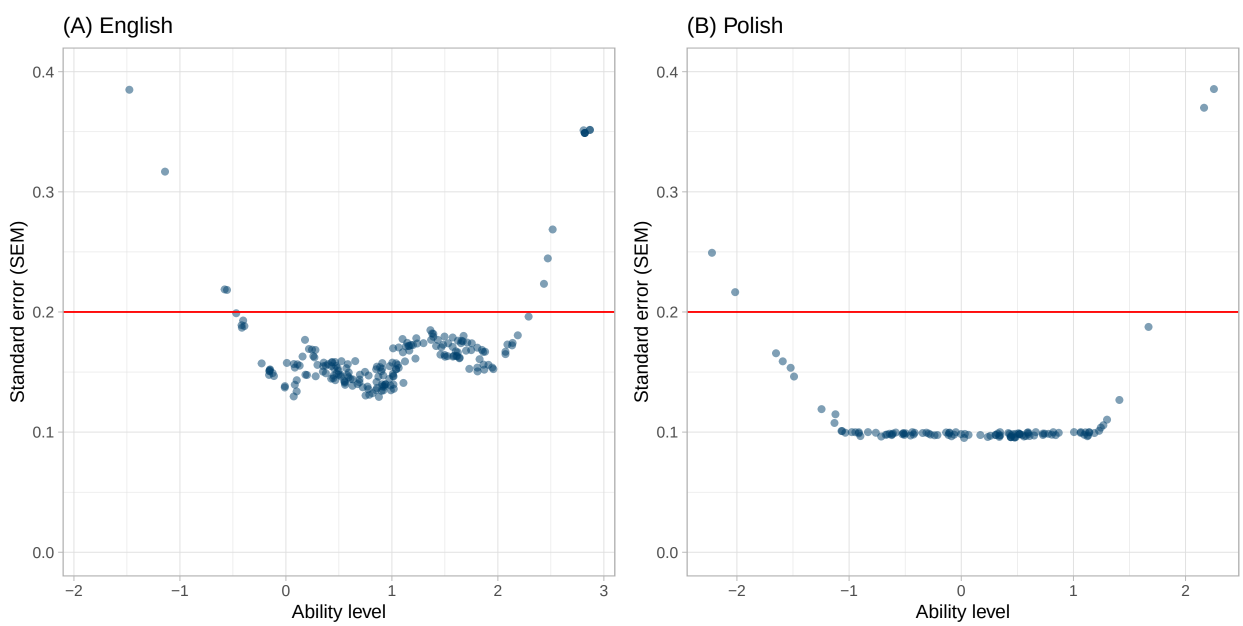


*Figure 4*. Standard Error of Measurement (SEM) by ability estimate in (A) English and (B) Polish. The red line marks the SE of 0.2.

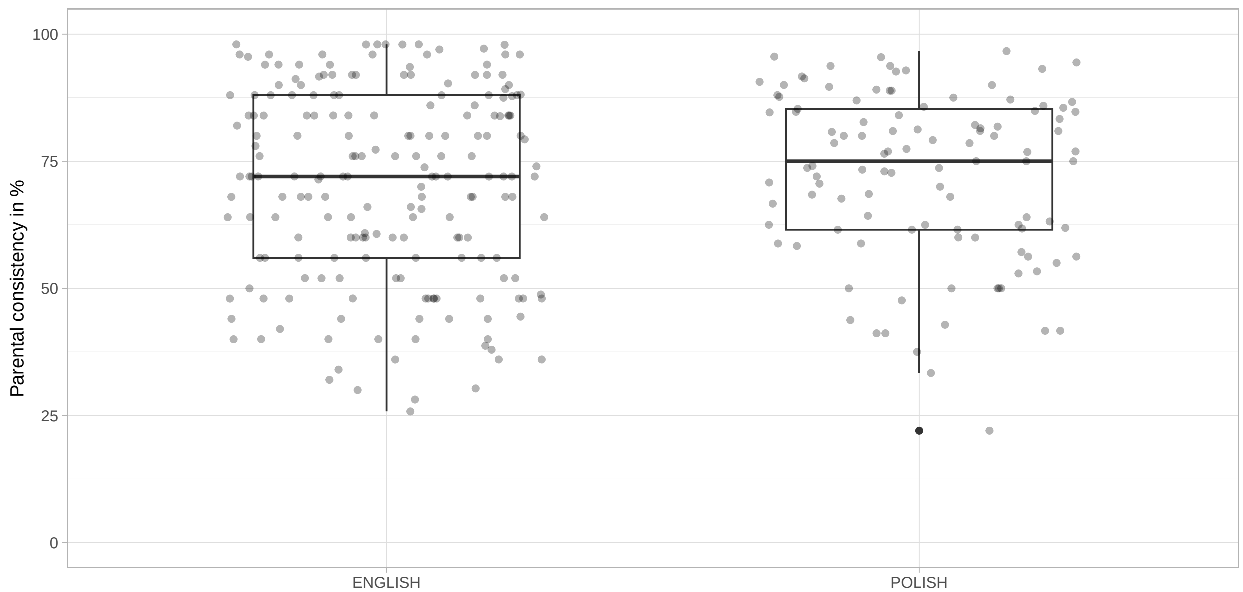


*Figure 5*. Percentage of parents' consistency in responding to items in the two CDI versions in the same way (in the English and in the Polish dataset).

## Parental consistency in responses

We compared the parental consistency in their responses in the full and CAT version of CDI in two ways - using a percentage of agreement and by calculating Cohen's Kappa, which additionally factors in chance agreement. When calculating the percentage agreement, for each parent we calculated the number of items to which they responded in the same way in both CDI versions against the whole number of items asked in both CDI versions. We found though the Median consistency was relatively high: 72% for English and 75% for Polish, it varied largely across parents in both samples (see Figure 5).

We also calculated Cohen's kappa per each parent, considering their observed agreement (between responses to the same items in full and CAT CDI) and expected agreement by chance. The kappa values indicated fair (or close to fair) agreement: English $M = 0.40$, $Mdn = 0.37$, Polish $M = 0.39$, $Mdn = 0.34$. *NOTE*: There's an Issue for that here

## Discussion

In points for now:

1. *Strong correlations with full CDI.*
   The CAT-CDIs were strongly correlated with the full CDI versions in both languages, which supports the overall validity of the CAT approach. These correlations held for both monolingual and multilingual participants. However, the Polish sample was small, so further research is needed—and planned—to confirm these findings.

2. *Cases of extreme divergence.*
   CAT-CDI scores diverged notably from full CDI estimates for only a small number of children in both language groups. In the Polish sample all divergent cases had low SEM, and their full CDIs were unusually short–maybe the full form underestimated their ability? In the English sample, their full CDIs were not very short, but SEM was relatively low, which is good.

3. *Item paramteres across languages.*
   Item parameters (i.e., difficulty and discrimination) showed moderate correlations across languages. These likely stem from variations in the calibration samples. Still, some semantic categories showed consistent patterns across languages, suggesting underlying structural similarities in lexical development.

4. *Item selection.*
   Each CAT version used c.a. half of the full CDI item pool. Few items were overused (mostly those with highest discrimination in the item pool or starting items). But we also saw items being chosen for CAT that wouldn't be considered very informative (i.e., of low discrimination). That's something to explore further, I wrote to Phil Chalmers (author of mirt and mirtCAT), maybe he'll be able to help?

5. *Efficiency and Reliability.*

CAT-CDIs were significantly shorter than the full CDIs while maintaining high reliability for most children. The few less reliable estimates were for children at the ends of the ability range (children with either very low or very high lexical skills). Comparing Polish and English samples, the ability ranges with low SEM differed (see Figure 3) - this again might be a result of the calibration samples and item parameters? This is a good place to stress the importance of well-balanced, diverse calibration datasets. Also, while the CAT-CDI is efficient and useful for screening or initial identification ("pomiar przesiewowy" in Polish), it may be less suited for diagnostic purposes where high precision is needed across the full ability range.

# References


Auguie, B. (2017). *gridExtra: Miscellaneous functions for "grid" graphics*. Retrieved from https://CRAN.R-project.org/package=gridExtra

Aust, F., & Barth, M. (2024). *papaja: Prepare reproducible APA journal articles with R Markdown*. https://doi.org/10.32614/CRAN.package.papaja

Barth, M. (2023). *tinylabels: Lightweight variable labels*. Retrieved from https://cran.r-project.org/package=tinylabels

Ben-Shachar, M. S., Lüdecke, D., & Makowski, D. (2020). effectsize: Estimation of effect size indices and standardized parameters. *Journal of Open Source Software*, *5*(56), 2815. https://doi.org/10.21105/joss.02815

Chalmers, R. P. (2012). mirt: A multidimensional item response theory package for the R environment. *Journal of Statistical Software*, *48*(6), 1–29. https://doi.org/10.18637/jss.v048.i06

Chalmers, R. P. (2016). Generating adaptive and non-adaptive test interfaces for multidimensional item response theory applications. *Journal of Statistical Software*, *71*(5), 1–39. https://doi.org/10.18637/jss.v071.i05

Chang, W., Cheng, J., Allaire, J., Sievert, C., Schloerke, B., Xie, Y., … Borges, B. (2024). *Shiny: Web application framework for r*. Retrieved from https://CRAN.R-project.org/package=shiny

Garnier, Simon, Ross, Noam, Rudis, Robert, … Cédric. (2023). *viridis(Lite) - colorblind-friendly color maps for r*. https://doi.org/10.5281/zenodo.4678327

Garnier, Simon, Ross, Noam, Rudis, Robert, … Cédric. (2024). *viridis(Lite) - colorblind-friendly color maps for r*. https://doi.org/10.5281/zenodo.4679423

Grolemund, G., & Wickham, H. (2011). Dates and times made easy with lubridate. *Journal of Statistical Software*, *40*(3), 1–25. Retrieved from https://www.jstatsoft.org/v40/i03/

Kassambara, A. (2023). *Ggpubr: 'ggplot2' based publication ready plots*. Retrieved from https://CRAN.R-project.org/package=ggpubr

Krajewski, G. (2025). *Multilada: MultiLADA's little helpers.* Retrieved from https://github.com/gkrajewski/Multilada

Müller, K. (2020). *Here: A simpler way to find your files.* Retrieved from https://CRAN.R-project.org/package=here

Müller, K., & Wickham, H. (2025). *Tibble: Simple data frames.* Retrieved from https://CRAN.R-project.org/package=tibble

R Core Team. (2025). *R: A language and environment for statistical computing.* Vienna, Austria: R Foundation for Statistical Computing. Retrieved from https://www.R-project.org/

Sarkar, D. (2008). *Lattice: Multivariate data visualization with r.* New York: Springer. Retrieved from http://lmdvr.r-forge.r-project.org

Slowikowski, K. (2024). *Ggrepel: Automatically position non-overlapping text labels with 'ggplot2'.* Retrieved from https://CRAN.R-project.org/package=ggrepel

Torchiano, M. (2020). *Effsize: Efficient effect size computation.* https://doi.org/10.5281/zenodo.1480624

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis.* Springer-Verlag New York. Retrieved from https://ggplot2.tidyverse.org

Wickham, H. (2023a). *Forcats: Tools for working with categorical variables (factors).* Retrieved from https://CRAN.R-project.org/package=forcats

Wickham, H. (2023b). *Stringr: Simple, consistent wrappers for common string operations.* Retrieved from https://CRAN.R-project.org/package=stringr

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., … Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/joss.01686

Wickham, H., François, R., Henry, L., Müller, K., & Vaughan, D. (2023). *Dplyr: A grammar of data manipulation.* Retrieved from https://CRAN.R-project.org/package=dplyr

Wickham, H., & Henry, L. (2025). *Purrr: Functional programming tools.* Retrieved from https://CRAN.R-project.org/package=purrr

Wickham, H., Hester, J., & Bryan, J. (2024). *Readr: Read rectangular text data.* Retrieved from https://CRAN.R-project.org/package=readr

Wickham, H., Vaughan, D., & Girlich, M. (2024). *Tidyr: Tidy messy data.* Retrieved from https://CRAN.R-project.org/package=tidyr

R (Version 4.4.3; R Core Team, 2025) and the R-packages *dplyr* (Version 1.1.4; Wickham, François, Henry, Müller, & Vaughan, 2023), *effectsize* (Version 1.0.1; Ben-Shachar, Lüdecke, & Makowski, 2020), *effsize* (Version 0.8.1; Torchiano, 2020), *forcats* (Version 1.0.0; Wickham, 2023a), *ggplot2* (Version 3.5.2; Wickham, 2016), *ggpubr* (Version 0.6.0; Kassambara, 2023), *ggrepel* (Version 0.9.6; Slowikowski, 2024), *gridExtra* (Version 2.3; Auguie, 2017), *here* (Version 1.0.1; Müller, 2020), *lattice* (Version 0.22.7; Sarkar, 2008), *lubridate* (Version 1.9.4; Grolemund & Wickham, 2011), *mirt* (Version 1.44.0; Chalmers, 2012, 2016), *mirtCAT* (Version 1.14; Chalmers, 2016), *Multilada* (Version 0.7.0; Krajewski, 2025), *papaja* (Version 0.1.3; Aust & Barth, 2024), *purrr* (Version 1.0.4; Wickham & Henry, 2025), *readr* (Version 2.1.5; Wickham, Hester, & Bryan, 2024), *shiny* (Version 1.10.0; Chang et al., 2024), *stringr* (Version 1.5.1; Wickham, 2023b), *tibble* (Version 3.3.0; Müller & Wickham, 2025), *tidyr* (Version 1.3.1; Wickham, Vaughan, & Girlich, 2024), *tidyverse* (Version 2.0.0; Wickham et al., 2019), *tinylabels* (Version 0.2.5; Barth, 2023), *viridis* (Garnier et al., 2023; Version 0.6.5; Garnier et al., 2024) and *viridisLite* (Version 0.4.2; Garnier et al., 2023)